# Warmup Is an Update Budget: Scheduler Semantics and First-Step Adam Impulses


## Abstract

Learning-rate warmup is meant to limit early optimizer motion, but implemented training loops often report only a nominal scalar schedule. The optimizer instead responds to the learning rate actually present at `optimizer.step()`, its internal state, and the resulting parameter displacement. We audit a concrete failure mode in scratch-trained Pre-LayerNorm Transformer classifiers: a scheduler-order mismatch lets the first Adam update execute at learning rate 1.0 before warmup-scale control is applied. At Adam's first bias-corrected step, active coordinates are approximately sign updates, so raw-gradient norm clipping does not by itself bound the displacement when the actual first-step learning rate is large. On AG News and DBpedia, this first-step impulse violates a safe update-ratio envelope, degrades accuracy and probabilistic metrics, and remains damaging in a single-bad-step ablation where all later updates follow the stable schedule. A calibrated update-ratio guard detects the violation; fail-fast stops invalid runs, and rescue replaces the unsafe update and recovers corrected-order performance. The practical lesson is that warmup should be audited as a realized update budget: report the nominal schedule, the optimizer-step learning rate, and normalized parameter displacement.


**Keywords:** learning-rate warmup, Adam, Transformer optimization, scheduler semantics, reproducibility, runtime guards

## 1. Introduction

Warmup is usually described as a learning-rate curve. In an implemented optimizer, however, the curve is only an intended control signal. The parameter trajectory is governed by the learning rate that is installed when the optimizer transition executes, by the optimizer state, and by the displacement that transition applies. A training run can therefore contain a reasonable nominal warmup formula while violating the intended early-update budget in parameter space.

This distinction is especially sharp for adaptive optimizers. In Adam, the first bias-corrected update has a coordinate-wise normalized form. On active coordinates, its magnitude is governed primarily by the actual first-step learning rate, not by the norm of the raw gradient. A scheduler-order mismatch that leaves a legacy learning rate of 1.0 in place for the first update can therefore create a large parameter-space impulse even when global gradient clipping is enabled.

We study this mechanism in scratch Pre-LayerNorm (Pre-LN) Transformer classifiers on AG News and DBpedia. The controlled design separates stable warmup, corrected scheduler order, legacy scheduler order without protection, fail-fast detection, rescue, and a single-bad-step causal ablation. The central measurement is not only the nominal schedule $\widetilde{\alpha}_t$, but the pair consisting of the actual optimizer-step learning rate $\alpha_t$ and the realized update-to-parameter ratio

$$r_t = \frac{\|\Delta\theta_t\|_2}{\|\theta_{t-1}\|_2 + \varepsilon_r}, \qquad \Delta\theta_t = \theta_t - \theta_{t-1}. \tag{1}$$

The ratio is not a universal stability theorem. It is an auditable measurement of whether a run respects the early displacement scale established by safe reference runs.

The contribution is broader than a bug report. The paper argues that warmup experiments should be specified in executable optimizer semantics: the schedule formula, the learning rate consumed by each optimizer transition, and the realized displacement. This perspective turns a hidden implementation mistake into a reproducible, measurable, and guardable failure mode.

**Contributions.** We make five specific contributions. First, we formalize four quantities that are often conflated: the nominal schedule, the learning rate before the optimizer step, the learning rate consumed by the optimizer, and the realized parameter displacement. Second, we give first-step Adam bounds explaining why an unsafe first optimizer-step learning rate can produce a sign-like displacement despite raw-gradient norm clipping. Third, we provide causal evidence that one unsafe first update is sufficient to damage training: the single-bad-step condition applies exactly one unsafe update and then follows the stable schedule. Fourth, we evaluate a calibrated update-ratio guard whose threshold is fixed using only safe stable/corrected runs. Fifth, we state a reporting standard for warmup reproducibility: log $\widetilde{\alpha}_t$, $\alpha_t$, and a normalized displacement such as Eq. (1).

## 2. Related Work

**Warmup as early-update control.** The original Transformer recipe used Adam with a warmup-and-decay schedule (Vaswani et al., 2017), and warmup became standard in large-batch and Transformer training (Goyal et al., 2017). Several analyses treat warmup as controlling early adaptive-optimizer motion: Ma and Yarats (2021) relate warmup to the magnitude of the Adam update term, Liu et al. (2020a) study variance in early adaptive learning rates, and Kosson et al. (2024) analyze warmup in GPT training through update-size measures. Our focus is procedural: even an appropriate nominal warmup cannot control early motion if the optimizer transition consumes the wrong learning rate.

**Transformer stability and normalization placement.** Transformer stability has been linked to normalization placement, residual amplification, initialization, and depth scaling. Xiong et al. (2020) show that Pre-LN changes initialization-time gradient behavior relative to Post-LN and can reduce warmup dependence in important regimes. Related stabilizers include normalization-centric modifications (Nguyen and Salazar, 2019), Admin initialization (Liu et al., 2020b), T-Fixup (Huang et al., 2020), and DeepNet scaling (Wang et al., 2022). These works address architectural and initialization mechanisms. We study a complementary implementation-level failure in which corrected Pre-LN runs train normally, but one unintended first optimizer update corrupts the trajectory.

**Adaptive optimizers, clipping, and relative updates.** Adam normalizes first-moment estimates by second-moment estimates (Kingma and Ba, 2015); its behavior has been connected to sign, magnitude, variance, and convergence issues (Balles and Hennig, 2018; Reddi et al., 2018). Explicit sign-based or sign-like optimizers make this connection direct (Bernstein et al., 2018; Chen et al., 2023). Decoupled weight decay changes the algebra by adding a separate learning-rate-scaled shrinkage term (Loshchilov and Hutter, 2019). Gradient clipping is a standard response to exploding gradients (Pascanu et al., 2013), but clipping the raw gradient is not the same as bounding Adam's parameter displacement. The update-to-parameter ratio is related in spirit to trust-region methods (Conn et al., 2000), Adafactor's update clipping and parameter-scale ideas (Shazeer and Stern, 2018), and layerwise adaptive scaling in LARS and LAMB (You et al., 2017, 2020); here it is instrumentation, not a new optimizer.

**Scheduler semantics, monitoring, and calibration.** Learning-rate schedulers are executable state machines. PyTorch documentation warns that optimizer/scheduler ordering changes the learning-rate value used by an update (PyTorch Contributors, 2026), and Hugging Face scheduler utilities expose warmup schedules as optimizer-coupled components (Hugging Face Contributors, 2026). Our guard follows a statistical-process-control pattern: estimate a safe envelope, then detect deviations online (Shewhart, 1931). This connects to broader work on ML testing, technical debt, and reproducibility (Breck et al., 2017; Sculley et al., 2015; Pineau et al., 2021). We report accuracy together with negative log-likelihood, expected calibration error, and Brier score (Guo et al., 2017; Brier, 1950). Deep double descent is a different phenomenon requiring evidence across model, data, or epoch complexity regimes (Nakkiran et al., 2020); our audited runs instead isolate a first-step implementation artifact.

## 3. Mechanism and Measurement

### 3.1. Nominal schedule versus executed optimizer input

**Definition 1 (Executed warmup quantities)** *At optimizer update $t$, let $\widetilde{\alpha}_t$ be the nominal learning rate prescribed by the schedule, $\alpha_t^{\text{pre}}$ the value stored in the optimizer immediately before the transition, $\alpha_t$ the value consumed by the optimizer transition, and $\alpha_t^{\text{post}}$ the value after any scheduler update. With gradient $g_t$ and optimizer state $s_{t-1}$,*

$$(\theta_t, s_t) = \Phi(\theta_{t-1}, g_t, s_{t-1}; \alpha_t), \qquad \Delta\theta_t = \theta_t - \theta_{t-1}. \tag{2}$$

*Warmup constrains the realized trajectory only when $\alpha_t$ matches the intended warmup-scale value at the moment of `optimizer.step()`.*

Definition 1 separates the schedule from the transition that changes parameters. A log containing only $\alpha_t^{\text{post}}$ can be misleading: the scheduler may write a safe value after an unsafe transition has already occurred. Eq. (1) measures the displacement caused by the transition itself.

**Proposition 2 (Learning-rate scaling for a fixed Adam first-step direction)** *Consider a fixed first-step Adam direction $u_1$ produced from a fixed clipped gradient and fixed optimizer hyperparameters, without weight decay. If the first update is $\Delta\theta_1(\alpha) = -\alpha u_1$, then for two positive learning rates $\alpha$ and $\alpha'$,*

$$r_1(\alpha') = \frac{\alpha'}{\alpha} r_1(\alpha). \tag{3}$$

*Thus a scheduler-order error that replaces a warmup-scale $\alpha_1$ by 1.0 scales the first relative displacement by the same factor, conditional on the same pre-step state and gradient.*

The proposition is elementary, but it is the quantity that warmup intends to control. If $\alpha_1$ is wrong, the relative displacement is wrong even if the schedule written after the step appears safe.

### 3.2. First-step Adam impulse

**Proposition 3 (First-step Adam update)** *Consider Adam without weight decay at the first optimizer update. Let* $g_1 \in \mathbb{R}^d$*,* $\beta_1, \beta_2 \in (0,1)$*, and* $\varepsilon_{\mathrm{Adam}} > 0$*. Adam computes*

$$m_1 = (1-\beta_1)g_1, \qquad v_1 = (1-\beta_2)(g_1 \odot g_1). \tag{4}$$

*After bias correction,* $\widehat{m}_1 = g_1$ *and* $\widehat{v}_1 = g_1 \odot g_1$*, so the first update is*

$$\Delta\theta_{1,i} = -\alpha_1 \frac{g_{1,i}}{|g_{1,i}| + \varepsilon_{\mathrm{Adam}}}. \tag{5}$$

*For active coordinates with* $|g_{1,i}| \gg \varepsilon_{\mathrm{Adam}}$*, the update is approximately* $-\alpha_1 \operatorname{sign}(g_{1,i})$*; if* $g_{1,i} = 0$*, the coordinate does not move.*

**Corollary 4 (Finite-epsilon active-coordinate bounds)** *Let* $\mathcal{A}$ *contain* $k$ *coordinates satisfying* $|g_{1,i}| \geq q\varepsilon_{\mathrm{Adam}}$ *for* $q > 0$*. Then, ignoring coordinates outside* $\mathcal{A}$*,*

$$\alpha_1 \frac{q}{q+1}\sqrt{k} \leq \|\Delta\theta_{1,\mathcal{A}}\|_2 \leq \alpha_1\sqrt{k}. \tag{6}$$

*When active coordinates dominate the update norm and* $q$ *is large,*

$$r_1 \approx \frac{\alpha_1\sqrt{k}}{\|\theta_0\|_2 + \varepsilon_r}. \tag{7}$$

**Lemma 5 (Clipping caveat)** *Suppose global norm clipping replaces* $g_1$ *by* $\bar{g}_1 = cg_1$ *with* $c > 0$*. For any coordinate satisfying* $|cg_{1,i}| \geq q\varepsilon_{\mathrm{Adam}}$*,*

$$|\Delta\theta_{1,i}| = \alpha_1 \frac{|cg_{1,i}|}{|cg_{1,i}| + \varepsilon_{\mathrm{Adam}}} \in \left[\alpha_1 \frac{q}{q+1},\, \alpha_1\right]. \tag{8}$$

*Thus clipping the raw gradient norm does not itself impose a small first-step Adam displacement when* $\alpha_1$ *is large and many coordinates remain outside the epsilon-dominated regime.*

These are optimizer-algebra statements, not a general convergence theorem. Epsilon-dominated coordinates, inactive coordinates, finite precision, initialization scale, parameter norms, and layerwise structure affect the realized ratio. Weight decay adds caveats: decoupled AdamW contributes a separate $-\alpha_1\lambda\theta_0$ term, while coupled $\ell_2$ regularization changes the effective gradient before normalization. For this reason the guard below is calibrated from measured safe updates rather than chosen from the nominal schedule alone. Full proofs are in Appendix A.

Table 1: Shared protocol for the audited Warmup Safety Gate experiments. The threshold is calibrated only from safe stable/corrected runs.

| Item | Value |
|---|---|
| Datasets | AG News, DBpedia |
| Models | Scratch Pre-LN Transformer classifiers: medium-4, large-6 |
| Seeds | 42, 123, 456 |
| Optimizer | Adam, $\beta_1 = 0.9$, $\beta_2 = 0.98$, $\varepsilon_{\text{Adam}} = 10^{-9}$ |
| Base LR / legacy initial LR | $10^{-4}$ / 1.0 |
| Warmup / clipping | 8000 steps / global norm 1.0 |
| Guard / diagnostic windows | first 100 / first 300 optimizer steps |
| Threshold rule | $\tau = 10 \max_{c \in \mathcal{C}_{\text{safe}}, t \leq 100} r_t(c)$ |

### 3.3. Calibrated update-ratio guard

Let $\mathcal{C}_{\text{safe}}$ be the safe calibration set containing stable warmup and corrected scheduler-order runs. For guard window $W = 100$, define

$$\tau = \kappa \max_{c \in \mathcal{C}_{\text{safe}}} \max_{1 \leq t \leq W} r_t(c), \qquad \kappa = 10. \tag{9}$$

The threshold is fixed before evaluating unsafe conditions. Fail-fast treats $r_t > \tau$ as a procedural invalidity and stops. Rescue replaces the unsafe update with a warmup-scale update and continues. The guard is therefore a protocol-specific runtime contract: it detects a measured violation of the safe early-update envelope, not a universal stability boundary.

## 4. Experimental Protocol

Experiments use AG News topic classification and DBpedia ontology classification with their standard train/test splits (Zhang et al., 2015; Auer et al., 2007). Models are scratch Pre-LN Transformer classifiers in two configurations, medium-4 and large-6. Conditions share seeds $\{42, 123, 456\}$, optimizer hyperparameters, warmup length, and gradient clipping. Table 1 summarizes the protocol; missing implementation details are treated as artifact limitations rather than filled in by assumption.

Every scheduler trace records LR before the step, LR used by the optimizer, and LR after scheduler update. We use $\alpha_t$ for the LR-used field. Update-ratio summaries are named by their measurement window: first-step ratio $r_1$, guard-window maximum $\max_{t \leq 100} r_t$, and early diagnostic traces over the first 300 steps. The reported quantitative evidence is based on calibrated full-run summaries; the current anonymized source package contains reconstructed summary CSVs for figure generation, not the original raw logs.

Table 2: Condition semantics. Fail-fast is a detector rather than a trained classifier; rescue is the trained intervention.

| Condition | Semantics |
|---|---|
| Stable | Stable warmup implementation; contributes to safe-envelope calibration. |
| Corrected | Corrected scheduler-order implementation; contributes to safe-envelope calibration. |
| Legacy/no guard | First update uses LR 1.0 and training continues without protection. |
| Fail-fast | Detector: detects the unsafe first update and stops; excluded from trained-model performance comparisons. |
| Rescue | Intervention: replaces the unsafe update with a warmup-scale update and continues training. |
| Single-bad | Causal ablation: applies one unsafe update, then follows the stable schedule. |

## 5. Results

### 5.1. The actual first optimizer-step LR identifies the artifact

Table 3 shows a representative first-two-step trace. Stable and corrected conditions use warmup-scale first updates. Legacy/no guard and single-bad use $\alpha_1 = 1.0$. Rescue starts from the same unsafe pre-step state as legacy but replaces the LR used by the first optimizer update.

Table 3: Representative first-two-step actual learning-rate trace from AG News, medium-4, seed 42. The decisive field is LR used: post-step LR can look safe after an unsafe update has already occurred.

| Condition | Step | LR before | LR used | LR after |
|---|---|---|---|---|
| Stable | 1 | 1.00e-04 | 1.25e-08 | 1.25e-08 |
| Stable | 2 | 1.25e-08 | 2.50e-08 | 2.50e-08 |
| Corrected | 1 | 1.00e+00 | 1.24e-07 | 1.24e-07 |
| Corrected | 2 | 1.24e-07 | 2.47e-07 | 2.47e-07 |
| Legacy/no guard | 1 | 1.00e+00 | 1.00e+00 | 1.24e-07 |
| Legacy/no guard | 2 | 1.24e-07 | 1.24e-07 | 2.47e-07 |
| Rescue | 1 | 1.00e+00 | 1.24e-07 | 1.24e-07 |
| Rescue | 2 | 1.24e-07 | 1.24e-07 | 2.47e-07 |
| Single-bad | 1 | 1.00e+00 | 1.00e+00 | 1.25e-08 |
| Single-bad | 2 | 1.00e+00 | 2.50e-08 | 2.50e-08 |
| Fail-fast | 1 | 1.00e+00 | 1.00e+00 | 1.24e-07 |

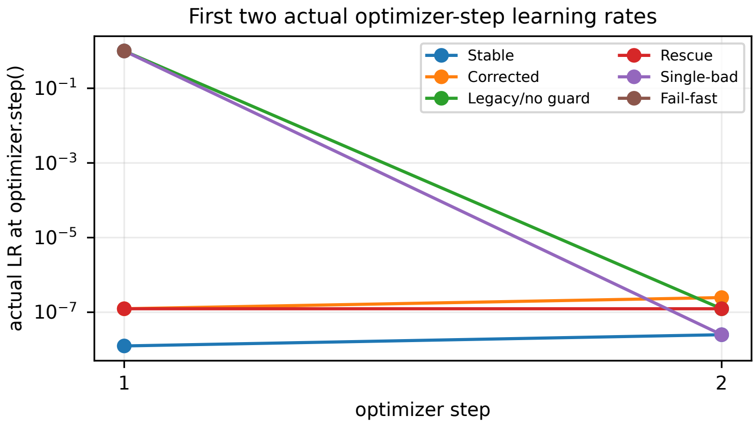


Figure 1: Actual optimizer-step learning rates for the first two steps. Legacy/no guard and single-bad apply the unsafe first update; rescue changes the learning rate consumed by that update.

### 5.2. The unsafe first update violates the calibrated update budget

The safe calibration maxima are 0.002004 for AG News and 0.001795 for DBpedia, yielding thresholds 0.020044 and 0.017947 under Eq. (9). Only stable and corrected runs contribute to these maxima. Figure 2 shows that unsafe first-step conditions lie far outside the envelope, while rescue returns to the warmup-scale region.

Table 4: Guard threshold calibration. Unsafe conditions are evaluated only after $\tau$ is fixed from safe runs.

| Dataset | Safe max $r$ | Multiplier | Threshold $\tau$ |
|---|---|---|---|
| AG News | 0.002004 | 10 | 0.020044 |
| DBpedia | 0.001795 | 10 | 0.017947 |

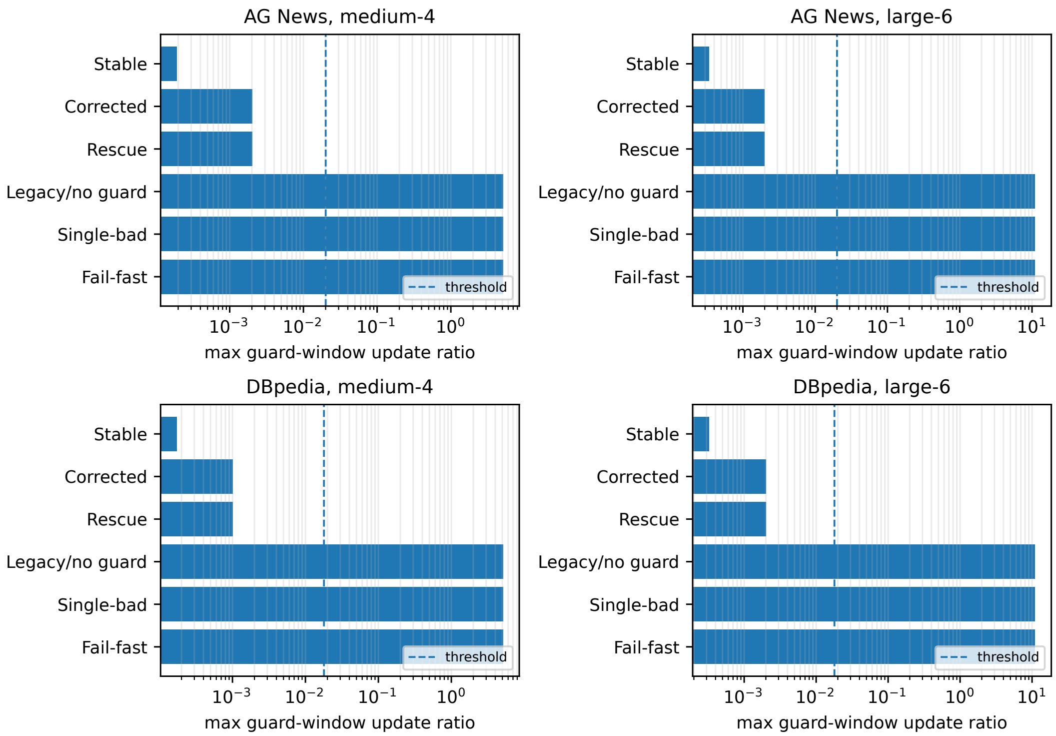


Figure 2: Maximum guard-window update-to-parameter ratios. Unsafe first-step conditions are separated from safe and rescued runs by a large margin; the dashed line is the dataset-specific threshold.

### 5.3. One unsafe first update is sufficient; rescue restores training

The single-bad condition applies the unsafe first update and then follows the stable schedule from step 2. Its persistent degradation isolates the early displacement as a causal factor in this protocol. Rescue performs the complementary intervention: it detects the unsafe update, replaces it with a warmup-scale update, and continues. Figure 3 shows that rescue recovers corrected-order performance, whereas single-bad remains degraded.

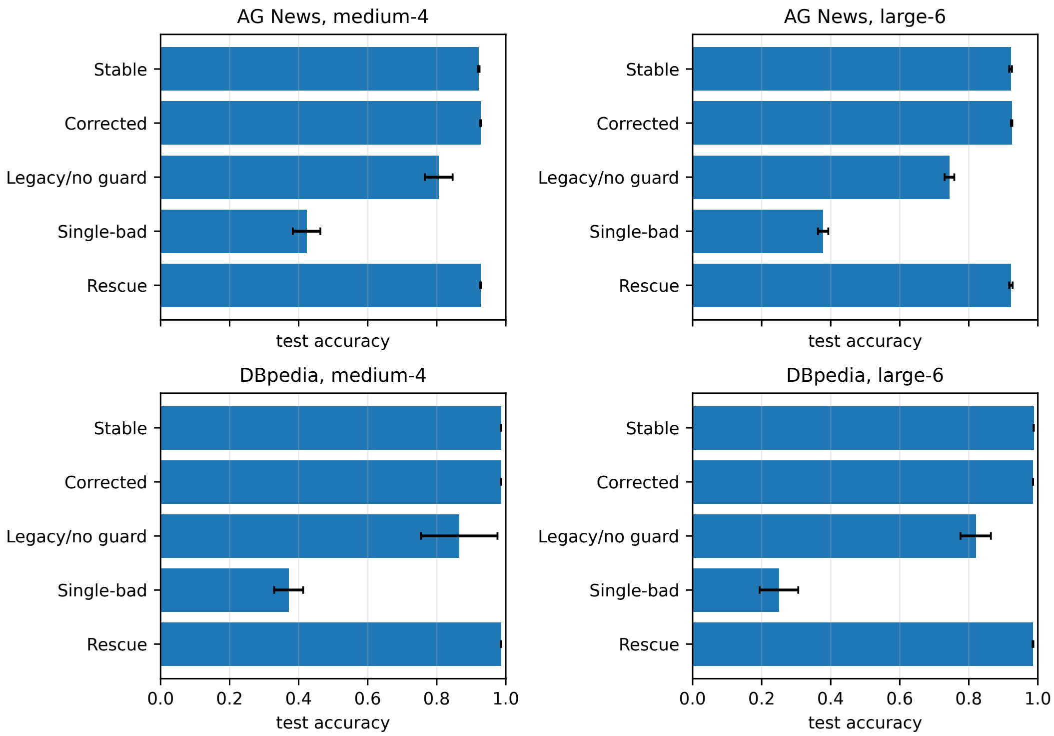


Figure 3: Test accuracy for trained conditions. Fail-fast is omitted because it stops rather than producing a trained classifier. Error bars show standard deviation over shared seeds.

Tables 5 and 6 report accuracy and descriptive probabilistic metrics. The same pattern appears in negative log-likelihood, ECE, and Brier score: unsafe first-step conditions degrade predictive quality, and rescue returns close to corrected-order training.

Table 5: AG News results. Values are means over shared seeds; max guard $r$ is the logged guard-window maximum.

| Model | Condition | Max guard $r$ | Acc. | ECE | NLL / Brier |
|---|---|---|---|---|---|
| medium-4 | Stable | 1.92e-04 | .922±.002 | .029 | .252 / .122 |
| medium-4 | Corrected | .002 | .927±.001 | .024 | .234 / .113 |
| medium-4 | Legacy/no guard | 5.091 | .806±.040 | .090 | .665 / .293 |
| medium-4 | Single-bad | 5.091 | .423±.040 | .575 | 15.578 / 1.150 |
| medium-4 | Rescue | .002 | .927±.001 | .025 | .241 / .115 |
| large-6 | Stable | 3.39e-04 | .921±.004 | .026 | .257 / .123 |
| large-6 | Corrected | .002 | .924±.002 | .034 | .262 / .119 |
| large-6 | Legacy/no guard | 10.998 | .744±.014 | .244 | 5.153 / .494 |
| large-6 | Single-bad | 10.998 | .378±.015 | .622 | 17.079 / 1.243 |
| large-6 | Rescue | .002 | .922±.005 | .030 | .259 / .122 |

Table 6: DBpedia results. Values are means over shared seeds; max guard $r$ is the logged guard-window maximum.

| Model | Condition | Max guard $r$ | Acc. | ECE | NLL / Brier |
|---|---|---|---|---|---|
| medium-4 | Stable | 1.73e-04 | .986±.000 | .006 | .057 / .022 |
| medium-4 | Corrected | .001 | .986±.000 | .005 | .054 / .022 |
| medium-4 | Legacy/no guard | 5.099 | .865±.111 | .081 | 1.030 / .215 |
| medium-4 | Single-bad | 5.099 | .371±.042 | .626 | 17.083 / 1.254 |
| medium-4 | Rescue | .001 | .986±.000 | .005 | .054 / .022 |
| large-6 | Stable | 3.24e-04 | .988±.000 | .006 | .051 / .019 |
| large-6 | Corrected | .002 | .986±.000 | .007 | .061 / .022 |
| large-6 | Legacy/no guard | 10.992 | .820±.044 | .175 | 4.156 / .352 |
| large-6 | Single-bad | 10.992 | .250±.056 | .749 | 20.668 / 1.499 |
| large-6 | Rescue | .002 | .986±.001 | .007 | .063 / .023 |

Because conditions share seeds, paired differences are the clearest effect summary. Legacy/no guard loses 12–18 accuracy points relative to corrected order on average across dataset/model groups; single-bad loses roughly 50–74 points relative to stable; rescue stays within a few tenths of a point of corrected order. Fail-fast and rescue both trigger on every unsafe run in the audited conditions; fail-fast stops, whereas rescue replaces the update and continues. These comparisons justify a causal claim about the first update in this protocol, not a universal claim about all warmup failures.

### 5.4. Capacity-only and double-descent explanations are not supported here

A capacity-only explanation would predict degradation under corrected update semantics. Instead, corrected and rescued runs train normally, while degradation tracks the unsafe first optimizer update. A gradient-explosion-only explanation is also incomplete because global norm clipping is enabled; Section 3 explains why clipping the raw gradient does not ensure a small adaptive-optimizer displacement. The evidence also does not support a deep-double-descent interpretation for these runs: the audited data do not establish a model/data/epoch complexity peak and instead isolate a procedural first-step artifact. This does not rule out double descent in other regimes.

## 6. Discussion, Limitations, and Reproducibility

The results support a narrow but important mechanism. Warmup failed because the first optimizer transition consumed the wrong learning rate. The trace in Table 3 shows that $\alpha_t^{\text{post}}$ can look warmup-scaled after the unsafe update has already been applied. The measured ratios in Figure 2 show that this is not a cosmetic logging discrepancy: the parameters moved far outside the safe early-update envelope.

The guard turns update scale into runtime validity instrumentation. Safe runs define a measured envelope; fail-fast and rescue give the training loop explicit semantics when a proposed update leaves that envelope. A run that violates the update budget at step 1 should not be interpreted as evidence about model depth, capacity, or dataset difficulty without first accounting for the procedural violation.

The empirical scope is intentionally limited: two text-classification datasets, two scratch Pre-LN Transformer configurations, Adam with one hyperparameter setting, and three shared seeds. The guard threshold is calibrated without leakage from unsafe conditions, but it is protocol-specific; $\kappa = 10$ should not be treated as a universal safe value. The reported logs contain first-step, guard-window, and early diagnostic-window ratios; full-training maximum update ratios and full-training layerwise ratios were not available and are not claimed. The probabilistic metrics are descriptive summaries. The anonymized source package contains reconstructed summary CSVs sufficient to regenerate the paper figures, not the original raw run logs. Before artifact evaluation, the release should add raw per-seed logs, complete scheduler traces, guard events, manifests, hardware/software versions, and the exact commit hash.

## 7. Conclusion

Warmup should be treated as an executed update budget, not only as a nominal learning-rate curve. In the audited scratch Transformer classifiers, a scheduler-order mismatch lets Adam take the first update at learning rate 1.0, creating a sign-like parameter-space impulse far outside the calibrated safe envelope. The single-bad-step and rescue interventions identify that first update as the damaging event in this protocol. Warmup experiments should therefore report the schedule formula, the learning rate actually consumed by each optimizer step, and normalized parameter displacement.

## References

Sören Auer, Christian Bizer, Georgi Kobilarov, Jens Lehmann, Richard Cyganiak, and Zachary Ives. DBpedia: A nucleus for a web of open data. In *The Semantic Web*, pages 722–735. Springer, 2007. doi: 10.1007/978-3-540-76298-0_52.

Lukas Balles and Philipp Hennig. Dissecting Adam: The sign, magnitude and variance of stochastic gradients. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 404–413, 2018. URL https://proceedings.mlr.press/v80/balles18a.html.

Jeremy Bernstein, Yu-Xiang Wang, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD: Compressed optimisation for non-convex problems. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 560–569, 2018. URL https://proceedings.mlr.press/v80/bernstein18a.html.

Eric Breck, Shanqing Cai, Eric Nielsen, Michael Salib, and D. Sculley. The ML test score: A rubric for ML production readiness and technical debt reduction. In *2017 IEEE International Conference on Big Data*, pages 1123–1132, 2017. doi: 10.1109/BigData.2017.8258038.

Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Hieu Pham, Xuanyi Dong, Thang Luong, Cho-Jui Hsieh, Yifeng Lu, and Quoc V. Le. Symbolic discovery of optimization algorithms. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/9a39b4925e35cf447ccba8757137d84f-Abstract-Conference.html.

Andrew R. Conn, Nicholas I. M. Gould, and Philippe L. Toint. *Trust Region Methods*. SIAM, 2000. doi: 10.1137/1.9780898719857.

Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch SGD: Training ImageNet in 1 hour, 2017. URL https://arxiv.org/abs/1706.02677.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

Xiao Shi Huang, Felipe Perez, Jimmy Ba, and Maksims Volkovs. Improving transformer optimization through better initialization. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 4475–4483, 2020. URL https://proceedings.mlr.press/v119/huang20f.html.

Hugging Face Contributors. Optimization: Schedules. https://huggingface.co/docs/transformers/en/main_classes/optimizer_schedules, 2026. Accessed 14 June 2026.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

Atli Kosson, Bettina Messmer, and Martin Jaggi. Analyzing and reducing the need for learning rate warmup in GPT training. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/059445c2d5b3ef918079851628fef1d6-Abstract-Conference.html.

Liyuan Liu, Haoming Jiang, Pengcheng He, Weizhu Chen, Xiaodong Liu, Jianfeng Gao, and Jiawei Han. On the variance of the adaptive learning rate and beyond. In *International Conference on Learning Representations*, 2020a. URL https://openreview.net/forum?id=rkgz2aEKDr.

Liyuan Liu, Xiaodong Liu, Jianfeng Gao, Weizhu Chen, and Jiawei Han. Understanding the difficulty of training transformers. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pages 5747–5763, 2020b. doi: 10.18653/v1/2020.emnlp-main.463.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

Jerry Ma and Denis Yarats. On the adequacy of untuned warmup for adaptive optimization. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(10):8828–8836, 2021. doi: 10.1609/aaai.v35i10.17069.

Preetum Nakkiran, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever. Deep double descent: Where bigger models and more data hurt. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=B1g5sA4twr.

Toan Q. Nguyen and Julian Salazar. Transformers without tears: Improving the normalization of self-attention. In *Proceedings of the 16th International Conference on Spoken Language Translation*, pages 193–202, 2019. doi: 10.18653/v1/2019.iwslt-1.17.

Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. On the difficulty of training recurrent neural networks. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 1310–1318, 2013. URL https://proceedings.mlr.press/v28/pascanu13.html.

Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d'Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research. *Journal of Machine Learning Research*, 22(164):1–20, 2021. URL https://www.jmlr.org/papers/v22/20-303.html.

PyTorch Contributors. How to adjust learning rate. https://pytorch.org/docs/stable/optim.html#how-to-adjust-learning-rate, 2026. Accessed 14 June 2026.

Sashank J. Reddi, Satyen Kale, and Sanjiv Kumar. On the convergence of Adam and beyond. In *International Conference on Learning Representations*, 2018. URL https://openreview.net/forum?id=ryQu7f-RZ.

D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips, Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, and Dan Dennison. Hidden technical debt in machine learning systems. In *Advances in Neural Information Processing Systems*, volume 28, 2015. URL https://papers.nips.cc/paper_files/paper/2015/hash/86df7dcfd896fcaf2674f757a2463eba-Abstract.html.

Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 4596–4604. PMLR, 2018. URL https://proceedings.mlr.press/v80/shazeer18a.html.

Walter A. Shewhart. *Economic Control of Quality of Manufactured Product*. D. Van Nostrand Company, 1931.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://papers.nips.cc/paper/7181-attention-is-all-you-need.

Hongyu Wang, Shuming Ma, Li Dong, Shaohan Huang, Dongdong Zhang, and Furu Wei. DeepNet: Scaling transformers to 1,000 layers, 2022. URL https://arxiv.org/abs/2203.00555.

Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 10524–10533, 2020. URL https://proceedings.mlr.press/v119/xiong20b.html.

Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks, 2017. URL https://arxiv.org/abs/1708.03888.

Yang You, Jing Li, Sashank Reddi, Jonathan Hseu, Sanjiv Kumar, Srinadh Bhojanapalli, Xiaodan Song, James Demmel, Kurt Keutzer, and Cho-Jui Hsieh. Large batch optimization for deep learning: Training BERT in 76 minutes. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/forum?id=Syx4wnEtvH.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28, 2015. URL https://papers.nips.cc/paper/5782-character-level-convolutional-networks-for-text-classification.

## Appendix A. Proofs

**Proof of Proposition 2.** By assumption, $\Delta\theta_1(\alpha) = -\alpha u_1$ and $\Delta\theta_1(\alpha') = -\alpha' u_1$ for the same vector $u_1$. Taking Euclidean norms gives $\|\Delta\theta_1(\alpha')\|_2 = (\alpha'/\alpha)\,\|\Delta\theta_1(\alpha)\|_2$. The denominator $\|\theta_0\|_2 + \varepsilon_r$ is fixed before the transition, so Eq. (3) follows.

**Proof of Proposition 3.** At $t = 1$, Adam's biased first and second moments are $m_1 = (1-\beta_1)g_1$ and $v_1 = (1-\beta_2)(g_1 \odot g_1)$. Bias correction gives $\widehat{m}_1 = m_1/(1-\beta_1) = g_1$ and $\widehat{v}_1 = v_1/(1-\beta_2) = g_1 \odot g_1$. Substitution into Adam's update $\theta_1 = \theta_0 - \alpha_1 \widehat{m}_1/(\sqrt{\widehat{v}_1} + \varepsilon_{\text{Adam}})$ yields Eq. (5) coordinate-wise. If $|g_{1,i}| \gg \varepsilon_{\text{Adam}}$, then $|g_{1,i}|/(|g_{1,i}| + \varepsilon_{\text{Adam}}) \approx 1$; if $g_{1,i} = 0$, the numerator is zero.

**Proof of Corollary 4.** For $i \in \mathcal{A}$, Proposition 3 gives $|\Delta\theta_{1,i}| = \alpha_1 |g_{1,i}|/(|g_{1,i}| + \varepsilon_{\text{Adam}})$. The condition $|g_{1,i}| \geq q\varepsilon_{\text{Adam}}$ implies $|g_{1,i}|/(|g_{1,i}| + \varepsilon_{\text{Adam}}) \geq q/(q+1)$, while the same fraction is at most 1. Summing squared magnitudes over $k$ active coordinates gives Eq. (6); Eq. (7) is the large-$q$ approximation when active coordinates dominate.

**Proof of Lemma 5.** After clipping, Adam's first-step formula is

$$\Delta\theta_{1,i} = -\alpha_1 \frac{c g_{1,i}}{|c g_{1,i}| + \varepsilon_{\text{Adam}}}.$$

The same finite-epsilon bound used in Corollary 4 gives Eq. (8). For large $q$, the multiplicative clipping factor cancels up to the additive epsilon term, leaving an approximately sign-like update.